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1. TRISO layers

- 10/10
- a) Kernel - the fissionable fuel
 - b) Graphite buffer - mechanically disengages kernel from rest of layers to accommodate swelling. Also absorbs fission recoils and fission gas due to porosity.
 - c) IPyC - inner pyrolytic carbon, puts SiC in compression and protects from fission products
 - d) SiC - pressure vessel layer, contains all fission products and gas, high mechanical strength
 - e) OPyC - outer pyrolytic carbon, puts SiC in compression, protects in manufacturing

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2. Carbide fuels produce less/re free oxygen, so no CO or CO₂ gas that will pressurize particle. Also good high temperature stability with graphite layers. Hard to keep from oxidizing in manufacturing.

Oxide fuels are easier to fabricate, since oxidation in high temperature manufacturing is allowed. But it will pressurize TRISO particle due to CO gas production.

- FP retention better in oxides

3. Fuel kernel fission product chemistry has five categories:

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- a) Soluble oxide (e.g. Y) - dissolves in cation sublattice
 - b) Insoluble oxide (e.g. Zr) - some solubility with anion sublattice
 - c) Metal (e.g. Pd) - forms precipitates at grain boundaries, not soluble
 - d) Volatile gas (e.g. I) - forms voids and can diffuse and react with layers
 - e) Noble gas (e.g. Xe) - forms voids and gas bubbles, stable

- Wanted $U \rightarrow UO_2$, FP carbides, oxygen potential

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4. Below 423K in SiC, swelling increases logarithmically with dose. At 423K, the SiC can amorphitize. Between 423-1073K, this is the point defect swelling regime. Black spot defects (interstitial clusters) accumulate, and the swelling increase logarithmically with dose until saturation is achieved. Saturation occurs sooner as irradiation temperature increases, leading to less swelling.

- don't conflate dose/swelling, sooner implies time/dose

4 cont.

After 1273 K (1000°C), we reach the void swelling regime. Swelling continues to go up with dose and does not saturate. This is due to Frank defect loops and the diffusion of vacancies into voids. The saturation in the point defect regime is due to the recombination of defects, which increase with temperature. However, voids continue to grow with dose.

9/10 5. Thermal conductivity in SiC decreases with irradiation, as interstitial clusters effectively scatter phonons. The concentration of point defects in the low T regime reach steady state, leading to saturation of the conductivity. At high T there is a linear proportionality between swelling and thermal conductivity at low T. - resistivity $\rightarrow \frac{1}{K_{th}}$
At high T, the thermal conductivity goes up slightly in the void swelling regime since voids are not as effective at scattering phonons, and point defects recombine. - think this is a slight misunderstanding
- voids don't impact K_{th} as much as interstitial clusters

18/18 6. Graphite shrinks initially under irradiation, before reaching a turnaround point at which it volumetrically increases. This is due to a few factors:

- Interstitials from irradiation form between basal planes as a kind of bridge, increasing the $\langle c \rangle$ direction
- Vacancies form in the basal plane, reducing the $\langle a \rangle$ direction
- Empty space from Mozowski cracks parallel to the basal plane accommodate the $\langle c \rangle$ -expansion, leading to an overall volumetric reduction.

Once the porosity/cracks are filled, then the graphite begins to swell with increasing dose.

7. TRISO Failures

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- a) Overpressurization - the production of fission gas and CO from fuel burnup pressurize the particle and put tension on the SiC barrier which can fail.
 - b) Fission product attack - Pd and Ag can diffuse from the kernel to the SiC and react to corrode the integrity of the pressure barrier. Cs can also diffuse and release through the SiC.
 - c) Irradiation-induced dimensional change of IPyC - irradiation can cause the graphite to shrink then expand, putting it in a complex stress state that must be relaxed by irradiation creep. If there is too much tension in the IPyC from shrinkage, it can crack and damage the SiC as a result. - Close enough, got there in the end

5/5 8. ZrC has been researched as an alternative to SiC due to its high temperature stability and resistance to Pd attack, as well as easier fabrication.

9/10 9. IHX must operate for long periods at extremely high temperatures (750-1000°C). They must avoid corrosion from the impurities (H_2O , CO, N_2). Primary candidates are nickel-based superalloys (e.g. 617) that are carbide precipitation-strengthened to avoid high T creep, and also form a protective chromium oxide to inhibit corrosion. There is a critical temperature at which the oxide will decompose, but that depends on the gas environment. Care must be taken to avoid decarburization, excessive carburization, or extreme corrosive oxidation. - wanted a little more.